

Bioimage analysis 37 years ago (1987)

- [NIH Image](#) is the only game in town (below \$150,000)
 - Written in Pascal for the Macintosh (Mac II) by Wayne S. Rasband
 - First released in 1987, shared on floppy disks
 - Supporting 8-bit 256 gray “colors”
 - If you wanted it, buy a Macintosh, or
- Windows supported through buggy closed-source C port *Scion Image*
 - Scion built frame-grabbers and wanted to expand to Windows
- Self-driving off-road in day and night time conditions demonstrated
- 1988: Yann LeCun publishes convolutional neural network [paper](#)
 - This is a technology, not a tool, few people notice, paper is fun to read today
 - Middle of second [AI winter](#) (1987-1993)



Bioimage analysis 27 years ago (1997)



- **ImageJ** (J for Java)---still the main game in town
 - Released in 1997 as rewrite of *NIH Image*
 - Apple in decline, the PC on the rise
 - 1995: Sun Microsystems created the Java programming language as an operating system–agnostic programming platform
 - Rasband wanted to support Windows, but not two sets of code
 - “*Instead of ‘write once, run everywhere’, Rasband found himself writing once and debugging everywhere.*”
- First autonomous coast-to-coast drive of the United States two years earlier